Automated Poster Generation Leveraging Multi-Agent Collaboration

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Abstract

We propose a multi-agent food poster generation system that converts user-provided keywords into high-resolution, publication-ready posters. Our pipeline comprises four stages: (1) Context Expander, which enriches minimal input using retrieval-augmented generation; (2) Visual Designer, performing text-to-image generation via Stable Diffusion 3; (3) Layout Planner, composing final posters with domain-specific templates and style presets; and (4) Quality Refiner, employing multi-task learning with MobileNetV2 to assess and improve visual quality based on key attributes (clarity, colorfulness, saturation). The system achieves good-quality results through coordinated multi-agent orchestration, demonstrating effective domain-specific poster generation with aesthetic coherence.

1. Introduction

In recent years, with the rapid advancement of generative artificial intelligence, particularly text-to-image models, automated poster design has become a focus of research and application. Various online tools, such as Canva's AI Poster Generator, allow users to produce visually appealing designs simply by providing a few textual keywords. Meanwhile, there are some papers about it. For example, Jin et al. (2022) [1] proposed a method that combines text-based image retrieval with layout optimization to automatically produce stylized posters.

Despite these advances, there remains a significant gap in domain-specific poster generation within the food and culinary domain. Existing generic tools often fail to capture the nuances of food aesthetics, composition of the culinary scene, and stylistic preferences (for example, dessert, retro, or menu-inspired designs). To address this problem, this project proposes a multi-agent food aesthetic poster generation system. Input: A set of user-provided keywords or textual descriptions (for example, "dessert," "retro," "cream cake," "hand-drawn style," "warm tone"). Output: A high-resolution poster image that meets aesthetic and thematic expectations for the given food-related prompt.

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2. Implementation

Our proposed agent pipeline is composed of four consecutive parts: Context Expander, Visual Designer, Layout Planner and Quality Refiner. The Context Expander takes user input with 3 to 5 words (for example, "Chocolate cake with candles on") and expands it into a more detailed and standard brief containing theme, emotional tone, color palette, and other relevant content to improve Visual Designer performance. The brief is taken as the prompt for Visual Designer which performs text-to-image generation accordingly. Its output becomes the principal part of the poster. The background of the poster is generated by the Layout Planner. It composes the main figure and the background according to aesthetic rules and outputs the intermediate poster, which is then scored to determine whether it becomes the final output. The scoring is carried out by the Quality Refiner. Its main duty is to score the intermediate poster based on the patterns learned by the scoring DNN and to give relevant refinement suggestions to Visual Designer if the scores did not pass a certain threshold (Figure 1). The organization task for the multi-agent system is powered by LangGraph.

2.1. Context Expander and Visual Designer

The quality of the prompt is an important factor for the picture generated by text-to-image models. Despite the importance, writing a proper prompt that can truly increase generation quality is not intuitive and often requires domain-specific skills [2]. The Context Expander can address deficiencies in terms of prompt engineering by embedding aesthetic descriptions in user input.

Our system utilized retrieval-augmented generation (RAG) for domain-specific knowledge. Aesthetic related documents are segmented using a document structure-based strategy, then vectorized and stored in the local vector database (FAISS). Once the user input is read into the system, it is vectorized and used to retrieve the top-k relevant chunks from the local vector database based on cosine similarity. The retrieved document chunks as well as user input are composed and sent to LLM for further contextual expansion. LLM then returns the expanded context in JSON format (Figure 2), which is optimal for data transmission in

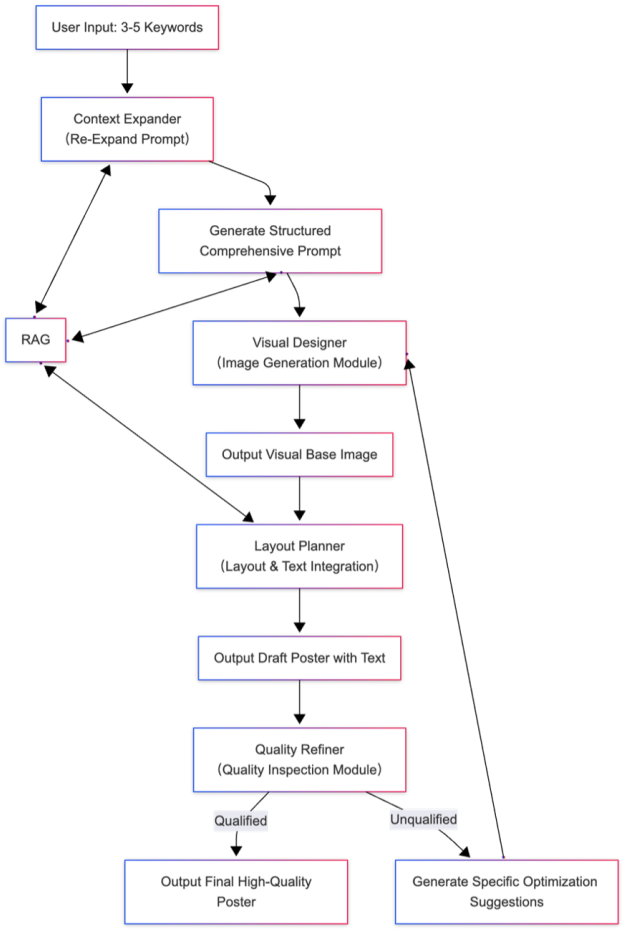


Figure 1. System Architecture

later processes.

For example, the user input is “chocolate cake with candles” and the LLM response is shown in Figure 2. It expanded the concept with more nuanced fields of the figure, such as style, tone, description, etc.

The Visual Designer agent takes the LLM response as input and calls Stable Diffusion 3 API for the text-to-image generation task. If the generated image has an aesthetic score below the threshold, the agent flow will return to Visual Designer stage (Figure 1) with suggestions embedded in the query to regenerate the principal figure.


```
{
  "theme": "Chocolate Cake with Candles",
  "style": "retro/vintage celebration",
  "tone": "warm and nostalgic",
  "emotion": "joyful indulgence",
  "tagline": "Sweet Celebration, Timeless Taste",
  "title": "Celebration Chocolate Cake",
  "description": "A rich, moist chocolate cake adorned with festive candles, perfect for marking special moments with warmth and sweetness.",
  "selling_points": ["Rich, Decadent Chocolate", "Perfect for Celebrations", "Homemade Comfort Aesthetic"],
  "cta": "Order for Your Celebration",
  "badge": "Celebration Special"
}
```


Figure 2. Context Expander Output

2.2. Layout Planner

The Layout Planner is responsible for creating the poster that consumers actually view. The Layout Planner takes over and turns this quality-assured material into a full $720 * 1280$ poster once the Visual Designer creates a food picture and the Quality Refiner confirms that the visual quality satisfies our aesthetic criteria. In other words, it serves as a link between ”good-looking content” and ”polished, ready-to-publish design.”

Three primary goals guide the Layout Planner’s design. First, we strive for professionalism and consistency: we employ template- and rule-based composition to guarantee stable structure, distinct hierarchy, and brand-like visual coherence across many posters rather than depending only on ad hoc or completely generative layouts.

Second, the module is designed with domain-specific aesthetics in mind. Our layouts and style presets are specifically tailored for the use case of food posters, which require attractive colors, a neat presentation, and easily accessible advertising content. Third, the module is designed to be extendable and modular, allowing the addition of new templates, styles, or copywriting techniques without having an impact on the system as a whole.

There are two primary inputs that the Layout Planner uses. The assessed food image created upstream, which is supplied as a file path, is the initial input. We can fairly conclude that the clarity, color balance, and general aesthetics of the image fall within an acceptable range because it has previously passed the Quality Refiner.

A structured creative brief in JSON format, which includes semantic data such as product type, topic, style, target audience, and call-to-action (CTA) text, is the second input. The module’s output is a $720 * 1280$ high-resolution PNG file.

A template-based layout structure forms the foundation of the Layout Planner. For food posters, we specify a number of built-in layout templates, including luxury, warm dessert, fresh drinks, minimalist, and vibrant.

In each design, a primary picture area, title and subtitle zones, body text sections for descriptions or sales points, and optional places for badges, pricing, or CTA buttons are encoded. We make sure that every poster maintains a clear visual hierarchy by separating the abstract layout structure from the real content: the product image is prominently shown, the main message is easily recognizable, and supplementary information is arranged in a predictable and understandable way.

The StylePlanner subsystem is in charge of translating semantic style requirements into specific visual characteristics on top of this structural framework. It examines the creative brief, focusing on elements such as food category (dessert versus drink), tone (warm versus fresh), and setting (celebration versus daily). It chooses a suitable style preset

linked to one of the templates based on these data. Every preset specifies font family, weight, color, and whether to utilize uppercase for various text roles (title, subtitle, body, CTA, badge). In addition, it outlines a color scheme with main, secondary, and accent hues that are adjusted for food aesthetics, such as using cooler, cleaner tones for drinks and warmer tones for sweets. This design enables us to methodically translate high-level style statements from the brief into useful visual criteria.

The Layout Planner incorporates an RAG-based design knowledge component in addition to StylePlanner. We have a modest design knowledge library that contains useful advice, including suggested contrast for readability, standard element spacing, or usual patterns for promotional layouts. Technology finds pertinent excerpts from the creative brief and uses them to improve template and style choices. For example, a guideline may recommend more delicate colors and more white space for expensive items, or sharper contrast and bolder writing for posters with a lot of discounts. Beyond what is recorded in templates alone, this incorporation of recovered design information assists the Layout Planner in aligning its outputs with accepted visual design standards.

There are several steps in the poster composition pipeline. The primary title, secondary tagline, product description, price details, and call-to-action phrases are the first copywriting pieces of the brief that the module creates by classifying the content into separate roles. Second, these text components are mapped to text blocks that are specified in the selected template; each block has alignment, typographic, and positional parameters. Third, ornamental elements that provide visual prominence without overpowering the product picture, such as badges, icons, or basic forms, are arranged according to the chosen style. The assessed food picture, text blocks, color scheme, and typographic arrangement are combined into a single $720 * 1280$ canvas by the PosterComposer component, which then creates the finished poster. As a result, a high-resolution PNG file is exported.

From an engineering standpoint, we focus on repeatability and robustness. It is simple to monitor and compare many runs since all file paths are resolved in relation to the project root, and outputs are regularly stored into a dedicated directory called outputs/posters. A different test script that feeds in example photos and mock briefs may also be used to independently test the Layout Planner. We can troubleshoot layout problems without executing the full multi-agent pipeline thanks to this modular approach. In addition, a basic input validation is performed. For instance, before beginning compilation, the module verifies that the input picture location exists and that the brief has the minimal necessary fields.

In conclusion, the Layout Planner uses a combination of template-based layout, style planning, and design knowledge retrieval to transform the upstream agents' work—semantically rich briefs and quality-assured images—into final, publication-ready posters. While maintaining enough flexibility to accommodate new designs and layouts in future extensions, it encodes domain-specific best practices for food posters into reusable templates and presets.

2.3. Quality Refiner

We propose a multi-task learning framework for food image aesthetics, combining overall scores with attribute subscores. Using a MobileNetV2 backbone and two-phase fine-tuning, the model predicts both holistic and granular qualities. Large-scale inference with KDE analysis identified clarity, color, and saturation as key determinants, enabling threshold-based diagnosis and actionable optimization suggestions. This pipeline delivers end-to-end assessment from quantitative scoring to practical guidance.

2.3.1. Data Preparation

The primary undertaking of this part is the establishment of a high-quality multi-task-labeled dataset. This dataset comprises 1000 food images, each meticulously annotated with two core dimensional labels: an overall aesthetic score and ten sub-scores for food attributes. These attributes include brightness, saturation, contrast, clarity, etc., collectively forming the Ground Truth required for model training. During the model training phase, these labels were utilized in supervised learning to guide the model to simultaneously acquire both holistic aesthetic perception and granular attribute decomposition capabilities.

2.3.2. Model Training

We adopted a Multi-Task Learning (MTL) architecture based on the MobileNetV2 backbone network (Figure 3). This model leverages the powerful feature extraction capabilities of MobileNetV2, feeding the extracted features into a shared layer, which then branches into two parallel output heads: one for predicting the overall aesthetic score and the other for predicting the ten attribute scores. All outputs are limited within the range of $[0, 1]$ using the Sigmoid activation function.

Following the architecture design, a Two-Phase Fine-Tuning strategy, based on the pre-trained MobileNetV2 backbone, was employed to efficiently migrate the model from general image recognition to the specific regression task of food aesthetic quality assessment.

The training process commenced with Phase A: Base Model Freezing and Top Layer Training. The objective of this phase was to rapidly adjust the weights of the newly added shared feature layer and the 11 multi-output heads (including the aesthetic score head and 10 attribute score heads). The model only updated the top-layer parameters,

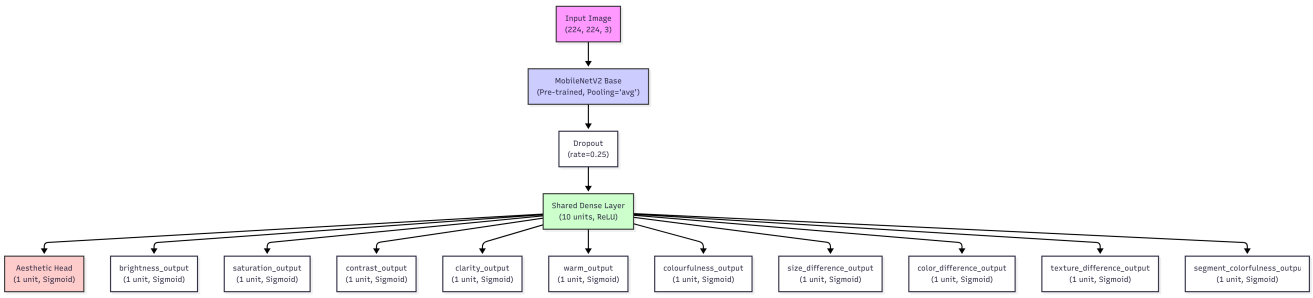


Figure 3. Model Architecture

utilizing a relatively high initial learning rate of 0.0001 and the Adam optimizer to achieve rapid weight convergence over a short period of 5 epochs.

Upon convergence of the top layers, training progressed to the more critical Phase B: Partial Backbone Unfreezing and Fine-Tuning. This phase aimed to achieve domain adaptation for the feature extractor. While the backbone was generally unfrozen, the code subsequently froze approximately 70% of MobileNetV2's lower-level weights to maintain the stability of fundamental generic features, allowing only the upper-level feature layers closer to the output to participate in training. The model was then recompiled using a significantly lower fine-tuning learning rate of 0.00001 and continued training. This extremely low learning rate was crucial to preventing the disruption of previously learned knowledge, ensuring that the model weights were precisely optimized for food aesthetic features in minute steps. This phase incorporated ModelCheckpoint monitoring the total loss on the validation set, ensuring that only the model weights with the best performance were saved, thereby completing the end-to-end optimization of the entire model.

2.3.3. Ablation Studies

To balance learning efficacy between primary task (aesthetic score) and auxiliary tasks (attribute scores), we conducted loss weight ablation experiments with various configurations. They have attribute loss weights of 0, 0.1, and 0.01 respectively. Ultimately, the loss weight for auxiliary tasks was set to 0.01 (with the aesthetic primary task weight at 1.0), ensuring the optimization focus remained on the overall aesthetic score.

In this task, the prediction accuracy of the overall aesthetic score is the primary measure of the diagnostic module's performance. Therefore, within the ModelCheckpoint settings, we selected the Mean Absolute Error (Figure 4) of the overall aesthetic score on the validation set as the criterion for saving the best model. Choosing this metric guarantees that the final deployed model is the version with the highest aesthetic perception accuracy, directly serving the module's quality assessment function. Monitoring the compounded loss might otherwise sacrifice the performance of primary task due to the optimization of auxiliary tasks.

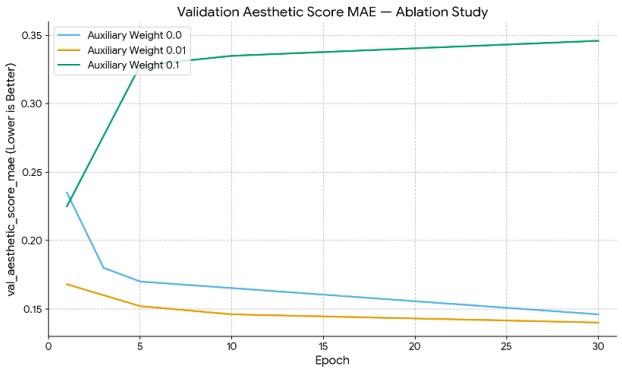


Figure 4. Ablation Experiment

2.3.4. Model Inference and Data Grouping

The trained model was used to perform large-scale inference and scoring on a dataset of 2000 images. Subsequently, by calculating the median of the aesthetic scores, the entire data set was divided into a High-Quality Group and a Low-Quality Group. This division served as the prerequisite for determining the diagnostic thresholds for the attributes.

2.3.5. Kernel Density Estimation and Attribute Diagnosis

Statistical analysis was performed on the High-Quality Group data to identify the characteristic attribute distribution of excellent images. We utilized the Kernel Density Estimation (KDE) method to estimate the Probability Density Function (PDF) for each attribute score. KDE, a non-parametric method, smoothly visualizes the true distribution shape of the two groups' score data, allowing for an intuitive comparison of the impact of different attributes on the aesthetic score.

The thresholds were first defined using the High-Quality Group. By calculating P5 and P95 of the high-quality score distribution(T_{max} and T_{min}), they established the rigid ideal tolerance interval for the diagnostic system. The

final diagnostic confirmation based on the distribution center of the low-quality group. By comparing the means of the two data groups, the script confirmed the primary mode of defect that causes low scores. Any attribute score falling outside of P5 and P95 exceeds the tolerance interval of 90% of the high-score samples, serving as a clear signal of aesthetic quality degradation. T_{min} was established as the critical point for attribute deficiency, and T_{max} was established as the critical point for attribute excess. If the mean of the low group is lower than the mean of the high group, attribute deficiency is judged to be the primary defect; conversely, excess is judged to be the primary defect.

Based on the distribution separability observed in the KDE plots (Figure 5) and an evaluation of the ability to act of experts, we ultimately filtered the 10 attributes down to three that were the most determinative of aesthetics, serving as the final measurement standards. They are clarity, $segment_{colorfulness}$ and saturation. The thresholds for these attributes were solidified, becoming the core parameters of the subsequent diagnostic module.

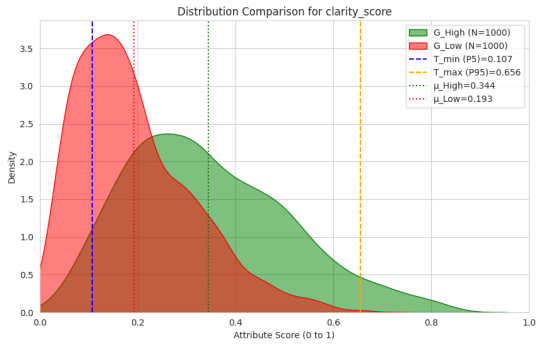


Figure 5. Clarity KDE

2.3.6. Quality Diagnosis and Suggestion Generation

The final stage involves quality diagnosis and suggestion generation, which is initiated by loading the trained model weights and the pre-computed thresholds. For every input image, the following process is executed.

The aesthetic attribute scores predicted for the input image are precisely compared against the statistical distribution of the high-quality image group to achieve accurate optimization. First, the model predicts the overall aesthetic score and the three key attribute scores. Second, in the evaluation phase, the overall aesthetic score is compared against the pre-set qualification threshold to quickly assess the image's viability. The third part is the diagnosis phase, where the predicted attribute scores are compared with the thresholds of the high-quality group to determine if the score falls outside the acceptable normal range. Finally, the suggestion phase follows the diagnosis, providing specific, actionable optimization advice accompanied by post-processing instructions, based on the attribute's score tendency. For instance, if the score is insufficient (below P5), the suggestion is to "increase" the corresponding parameter; if the score is excessive (above P95), the suggestion is to "decrease" the parameter. If the overall aesthetic score is unqualified but all key attribute scores fall within the P5-P95 range, the advice is to adjust the parameters towards the mean of the high-quality group to enhance overall aesthetic appeal.

Ultimately, this module returns the overall aesthetic score and structured optimization suggestions, completing the cycle from quantitative assessment to actionable guidance.

3. Discussion

According to the system log, for each iteration, the average execution time for Concept Expander, Visual Designer, Layout Planner, and Quality Refiner is approximately $21s$, $20s$, $0.2s$, and $0.7s$ correspondingly. And the add up time for each iteration is around $42s$. As a result, the optimization of Concept Expander and Visual Designer is key to system responding time improvement. In future system optimization, we will implement semantic caching for both Concept Expander as well as Visual Designer results so that the system will return cached results for semantically similar user inputs. The user can choose between fast and slow versions, and the system will return a cached or newly generated poster correspondingly.

References

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